# Does Geography Influence Prosperity?

A Macroeconomic Analysis of GDP vs. Geographic Factors

Target Audience: Global Policy Analysts & Development Agencies

# 1. Motivation & Policy Rationale

## The Core Question

To what extent does geographic placement—specifically distance from the equator—associate with a nation's economic prosperity and human development?

## Target Audience

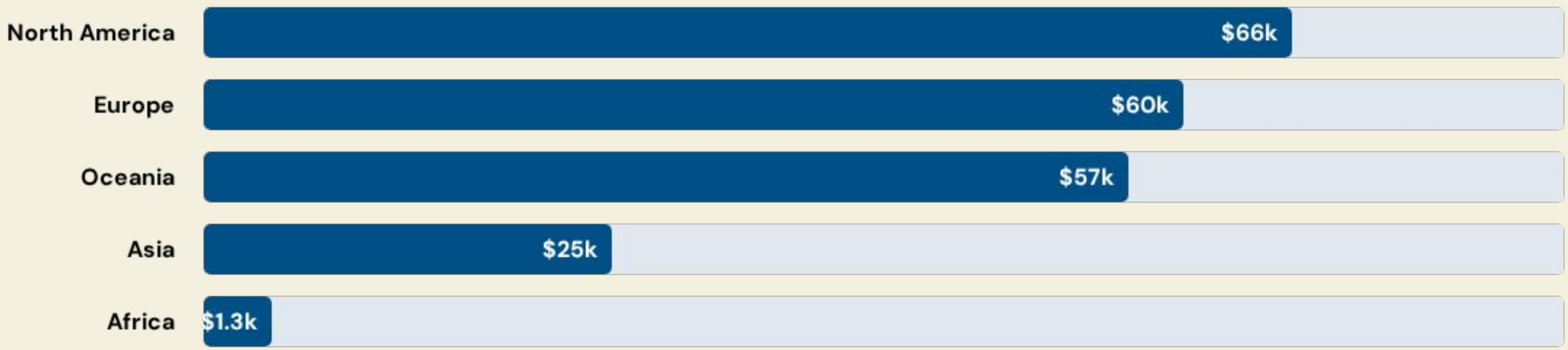
- **International Development Agencies:** To refine allocation of aid based on structural constraints.
- **Economic Policy Analysts:** To understand "latitude hurdles" vs. institutional quality.

# 2. Data Sourcing & SQL Integration

| Process | Logical Objective | Implementation Highlights |
|---|---|---|
| **Data Sourcing** | Cross-domain validation | Merged World Bank GDP API with UN HDI & Spatial Attributes. |
| **SQL Strategy** | Multi-table Relational Integrity | Complex JOINs on ISO codes to unite geography with metrics. |
| **Quality Checks** | Integrity Maintenance | Evaluated missing territorial data and outlier anomalies. |

*"We used complex SQL aggregations to cleanly map markers, filtering out invalid records to ensure a high-integrity foundational dataset."*

# 3. Insights: Regional Economic Disparity




## Key Takeaway

Visual analysis reveals a distinct macroeconomic "floor" and "ceiling" based on regional geography, confirming that land-access and latitude are strong growth indicators.

# 4. Bridging to Modeling: Engineered Features

### Absolute Latitude

Neutralized Northern/Southern hemisphere bias to measure pure distance from the equator as an economic friction variable.

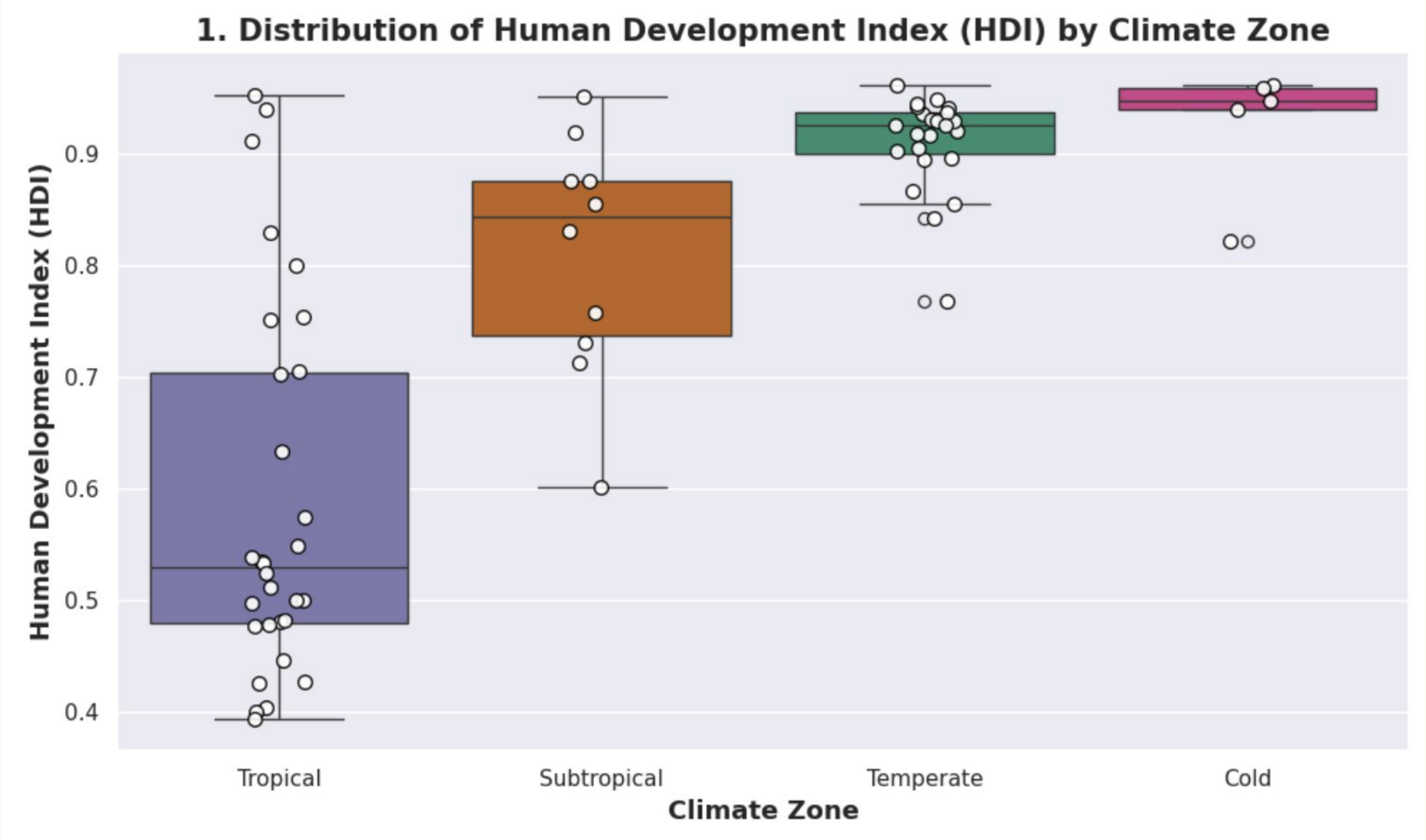
### Climate Zones

Mapped continuous coordinates to 4 distinct zones (Tropical to Cold) to capture environmental constraints on labor and industry.

### Standard Scaling

Normalized thousand-dollar GDP values with double-digit coordinates so the model doesn't over-weight the numeric scale.

# 5. Results:



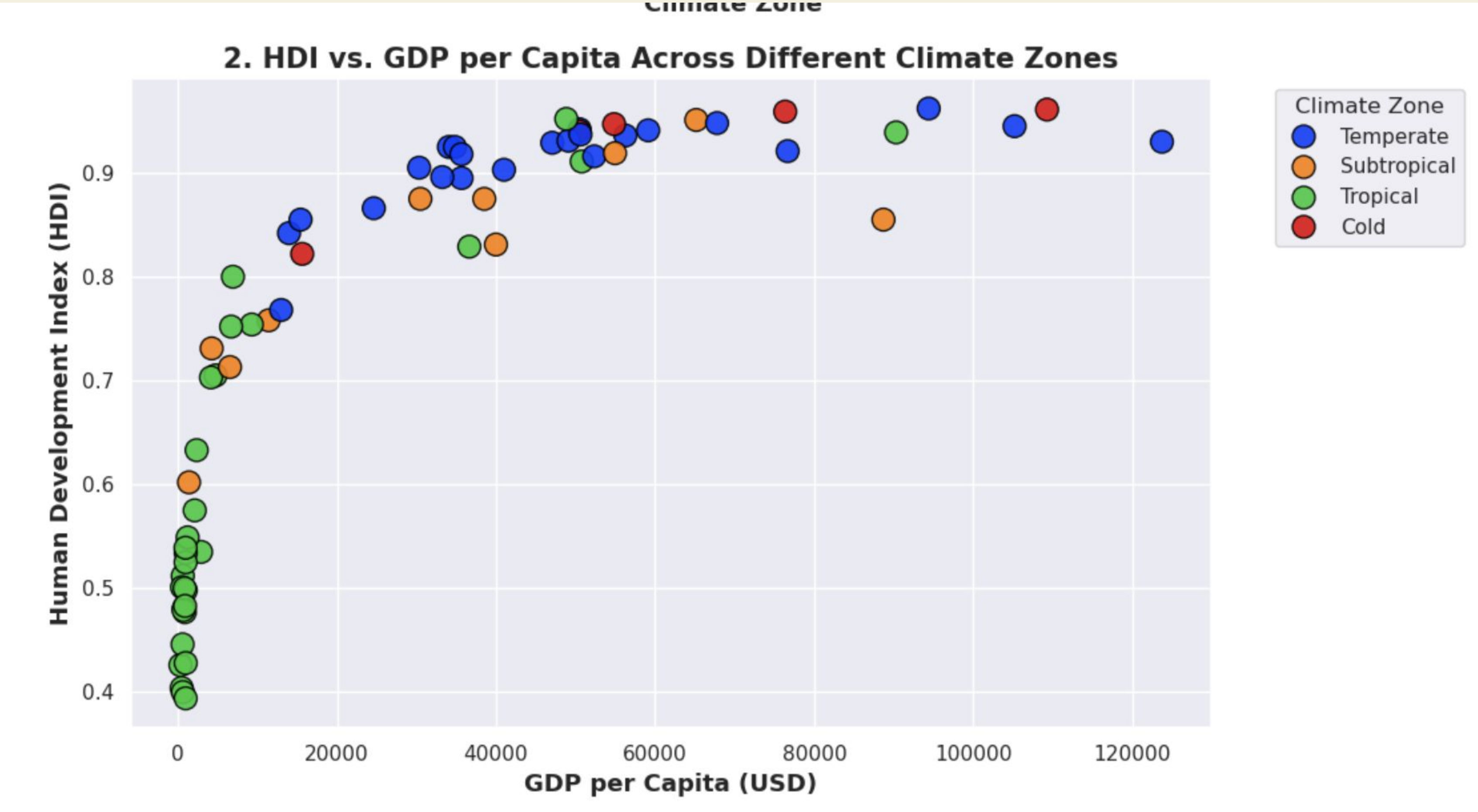

**The "Climate-Development" Gap:**

- Stark contrast between Tropical zones - rest of the world
- Massive gap between median Tropical HDI - median Cold HDI

**Predictability and Stability:**

- Temperate, Cold climates: high HDI is almost guaranteed
- Tropical climates: massive inequality

# 5. Results:

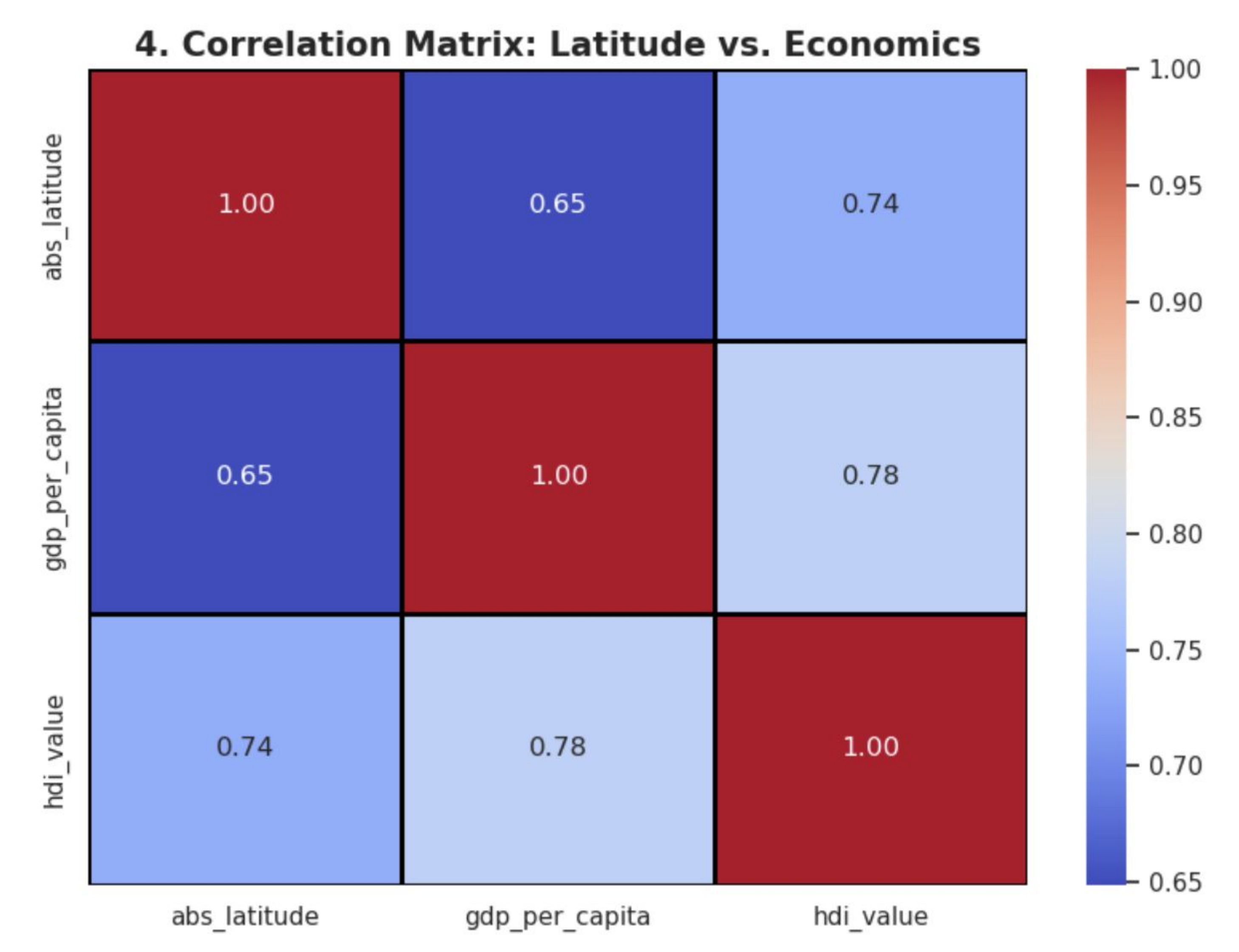



**Diminishing Returns:** HDI rises sharply with initial economic growth but plateaus significantly once a nation's GDP per capita exceeds roughly $40,000 USD.

**Tropical Clustering:** Countries located in tropical climate zones (green) are heavily concentrated at the lowest levels of both GDP per capita and human development.

**Temperate Dominance:** Countries in temperate zones (blue) dominate the high-income spectrum, achieving and sustaining the highest overall HDI levels.

# 5. Results:




**Strong Global Development Alignment:** There is a strong positive correlation (0.78) between GDP per capita and the Human Development Index (HDI), showing that economic output closely aligns with broader human welfare metrics.

**Geography Linked to Wealth:** Absolute latitude shares a moderate-to-strong positive correlation with both GDP per capita (0.65) and HDI (0.74), indicating that countries farther from the equator tend to have higher economic and development scores.

**Perfect Self-Correlation:** The diagonal values of 1.00 simply represent the perfect baseline correlation of each distinct variable matching with itself.

# 6. Predictive Modeling Results

0.66

**R-Squared Score**

## Interpreting the Model

Our model explains 66% of the variance in global GDP per capita using only geography and basic development markers.

**The "Why":** We chose an explanatory model because policy makers need to see the *weight* of each factor. This demonstrates that while geography is a powerful constraint, the remaining 34% represents the vital role of human institutions and policy.

# 7. The Verdict & Next Steps

## Final Conclusion

Geography acts as one of the primary gears in the global economic machine. However, it is an association, not absolute destiny.

## Future Work

If granted more resources, we will expand the model to include:

Institutional Quality (Rule of Law)
Education Attainment Rates
Global Trade Openness